

Procedure, Not Causal Vocabulary: A Matched-Access Test of Interventional Reasoning in Foundation Models

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Abstract

Prompted interventional gains are often read as evidence of causal reasoning. Our matched test identifies a different mechanism: procedure, not causal vocabulary, drives the gain. We hold information, examples, and a complete clamp-and-propagate algorithm fixed while changing only seven causal phrases. Across seven models from three providers on linear-Gaussian structural causal models, the neutral procedure improves Observational-Interventional Divergence (OID) by +0.220 over naive prompting ($n=70$ matched cells, $k=30$ SCM clusters, 95% CI [+0.111,+0.346], Romano-Wolf adjusted $p=0.003$) and repairs 28 of 70 cells. Recasting the same algorithm in causal language contributes -0.051 additional improvement (95% CI [-0.125,+0.017]); every model-specific interval crosses zero, and formal equivalence holds at the pre-specified ± 0.12 margin. On LaLonde/NSW, the procedure closes 73% of the observational confounding gap (95% interval [59%,81%]). Direct numerical methods remain more accurate, revealing the right system boundary: models can express an intervention, but the computation should remain explicit. A $do()$ -prompt cannot establish causal understanding unless it beats the same algorithm stated neutrally.

Introduction

A $do()$ -prompt is not a test of causal understanding when it also tells the model exactly how to compute the answer. Systems that forecast consequences under actions require the interventional quantity $\Pr(Y \mid do(X=x))$, which differs from $\Pr(Y \mid X=x)$ when common causes induce observational association [3]. Foundation models can be prompted to “apply the do-operator,” but ordinary prompt comparisons confound causal semantics with the algorithm embedded in the prompt. A better answer may demonstrate causal understanding, or it may demonstrate competent execution of supplied steps. Those are different scientific claims.

This distinction matters directly for temporal simulation. Each rollout repeatedly clamps an action or policy variable, preserves the current state where appropriate, and propagates the change forward. If an explicit recipe produces the improvement, then reliable systems should expose and verify that recipe rather than treating causal terminology as a capability signal. We first isolate this single-step primitive, then examine how scaffolding behaves as rollout horizons grow.

Recent benchmarks test causal explanation, scientific analysis, interventions, and counterfactuals in increasingly realistic settings [6–11]. They show where models succeed or fail across full causal-analysis pipelines, study design, and the causal lad-

der. We ask an orthogonal attribution question: if the correct graph, estimand, and computation are already supplied, does causal vocabulary improve execution beyond matched neutral instructions?

We use matched-access prompts to distinguish these mechanisms. The causal-language and neutral procedural prompts contain the same four ordered steps, worked example, precision instruction, and information. They differ by seven phrase substitutions, such as “intervene on” versus “fix” and “causal order” versus “listed order”; the causal prompt is seven tokens longer. The key comparison is not naive versus causal wording. It is naive versus the neutral procedure, followed by neutral procedure versus the same procedure expressed causally.

The result is unambiguous. The neutral procedure sharply reduces interventional error for every evaluated model. Replacing neutral terms with causal vocabulary adds no detectable improvement: the pooled interval and all seven model-specific intervals cross zero. Formal equivalence holds at the pre-specified ± 0.12 margin, with tighter margins reported separately. Provider, threshold, and missingness analyses preserve the procedural gain.

This paper contributes an attribution test that causal-prompting evaluations currently lack. Matched access holds the information and algorithm fixed and varies only the terminology. Component ablations show that clamping, ordering, and descendant propagation recover nearly all of the gain. LaLonde/NSW, nonlinear SCMs, causal benchmarks, counterfactual pairs, and temporal rollouts then stress-test where the result travels [17, 18]. The central question is no longer whether causal prompting helps. It is what, exactly, produced the improvement.

Separating procedure from terminology

Tasks and estimand. We generate six-node SCMs $x_j = f_j(\sum_{i < j} W_{ij}x_i) + \varepsilon_j$ with known DAGs. The primary family is linear-Gaussian, for which the intervention target is analytic. Additional families are standard nonlinear, discrete, and a purpose-built confounded nonlinear family. For query $(X=x, Y)$, we report

$$\text{OID} = \left| \hat{Y}_{\text{model}} - \mathbb{E}[Y \mid do(X=x)] \right|.$$

The graph and equations are supplied for the matched endpoint, isolating execution from graph recovery. When structure is instead elicited, only 1.1% of 282 inferred graphs are exact. We therefore treat graph discovery as a separate problem rather than folding it into the main score.

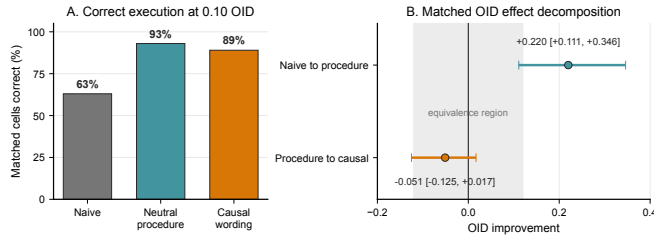


Figure 1: Matched access separates procedure from terminology. A: With graph and equations supplied, the neutral procedure increases cells within 0.10 OID from 44/70 to 65/70; causal wording reaches 62/70. B: The procedure yields a positive paired OID effect, while the causal-vocabulary increment crosses zero. The shaded region is the prespecified TOST margin; the plotted intervals are SCM-clustered 95% CIs, while formal TOST uses the corresponding 90% interval.

Prompt arms. The naive arm asks for the outcome under the stated value. The procedural arm instructs the model to fix the target, preserve non-descendants, propagate through descendants in listed order, and report the result. The causal-language arm gives exactly the same procedure using *do()*, intervention, and structural equation terminology. Both prompts have the same example and four common steps, and are nearly identical in length. Appendix prints the canonical instruction blocks, exhaustive phrase diff, exact model endpoints, and invocation controls. A deterministic implementation provides an execution ceiling. A data-matched OLS comparator sees the same 12 observational rows as the weights-hidden LLM arm.

Analysis. Our main comparison is the paired OID improvement from naive to procedural prompting on linear SCMs. We also test whether the procedural and causal-language arms are practically equivalent. Intervals resample SCMs, the independent unit, while keeping all model observations within each SCM. We use Romano-Wolf stepdown correction within the primary and secondary improvement families [21]. Equivalence is tested at the prespecified ± 0.12 OID margin, near the 0.10 design MDE at 80% power, with tighter margins reported as sensitivity analyses. Seven models from three providers contribute 70 evaluable cells over 30 SCMs. A cost-constrained incomplete-block schedule assigns endpoint-specific subsets of the 30 SCMs; one failed cell was replaced, yielding 71 attempts and 70 complete pairs. Structured parse failures remain missing rather than being silently regenerated.

Results

The supplied algorithm repairs interventional execution. Mean OID falls from 0.351 under naive prompting to 0.131 under the neutral procedure, a gain of +0.220 (95% CI [+0.111, +0.346], adjusted $p=0.003$). Median OID falls from 0.247 to 0.042. At the prespecified 0.10 threshold, correct cells rise from 44/70 to 65/70 (Fig. 1). The error taxonomy gives the same mechanistic account: 28/70 cells move from estimand-selection or arithmetic failure into the correct category.

Causal vocabulary contributes nothing measurable beyond the algorithm. Mean OID is 0.131 for the neutral procedure and 0.182 for its causal-language twin. The paired causal-

Procedural scaffolding is robust across components, thresholds, models, and missingness

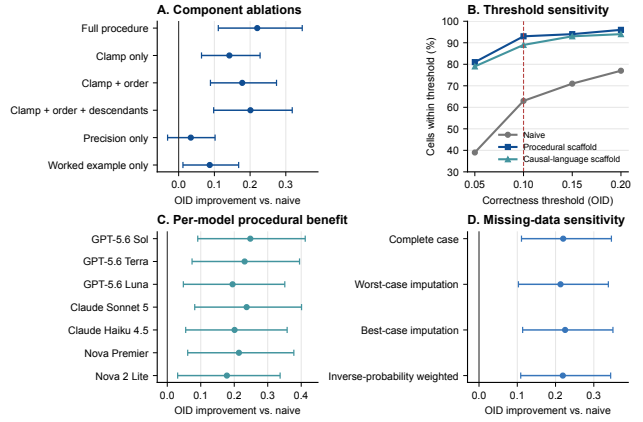


Figure 2: Robustness and mechanism. Clamping is the largest component; the conclusion persists across thresholds, all evaluable models, and missing-data analyses. Error bars are 95% intervals.

vocabulary increment is -0.051 (95% CI [-0.125, +0.017]). Its 90% interval, [-0.117, +0.009], lies inside the prespecified ± 0.12 equivalence bounds, although equivalence is inconclusive at margins of ± 0.10 or tighter. Every model has a positive naive-to-procedural estimate with a confidence interval excluding zero, while every procedural-to-causal interval crosses zero. The contrast is consistent across the complete model panel: the algorithm matters; causal terminology does not measurably improve it. Formal equivalence at tighter margins remains a sensitivity question.

The mechanism is clamp, order, and propagate. Clamping alone recovers 65% of the full observed scaffold gain (Fig. 2A). Adding variable ordering and descendant selection raises the recovered share to 91%. These cumulative ablations are not an additive decomposition because instructions may interact; precision instructions alone are inconclusive. At OID thresholds from 0.05 to 0.20, the procedural and causal-language arms differ by at most three of 70 cells. Complete-case, worst-case imputation, best-case imputation, and inverse-probability weighting yield procedural effects between +0.213 and +0.225. A deterministic script achieves zero OID, and data-matched OLS reaches 0.089. Once the equations are executable, hiding the computation inside an LLM adds error rather than capability.

The algorithm transfers beyond the primary synthetic family. For LaLonde/NSW, the randomized experimental ATE is \$1794 (SE \$632, 95% CI [\$555, 3033]) [24, 25]. Relative to this target, the observational construction has OID \$10279 and the adjusted scaffold has OID \$2745. The reduction is \$7535 (95% CI [\$7163, 7875]), closing 73% of the confounding gap with interval [59%, 81%] (adjusted $p=0.001$). OLS remains better by \$1691 (95% CI [1391, 2024]), which bounds the practical claim. The LLM receives a 12-stratum summary, supplied DAG, adjustment set, and estimand; Appendix gives the reconstruction protocol and calibration arm.

The purpose-built confounded nonlinear family also improves (+0.107, 95% CI [+0.036, +0.177], adjusted $p=0.041$).

Standard nonlinear and discrete families are inconclusive after correction. Corr2Cause improves before correction but not after it ($p=0.113$); CLADDER and Twins are null. The strongest gains therefore appear where the supplied clamp-and-propagate operation directly addresses confounding. The broader results identify a mechanism and its operating range, not a generic prompting effect.

Implications for evaluation and systems

Matched access changes how causal-prompting results should be interpreted. InterveneBench, CauSciBench, and CausalT5k test broad causal capabilities [7–9]. Our control asks what produced the gain. When the same algorithm works in neutral language, the improvement demonstrates execution of a supplied procedure, not an additional contribution from causal vocabulary. Evaluations should therefore pair causal prompts with matched neutral controls before attributing performance to causal understanding.

The same result suggests a practical architecture. Language models can translate questions into explicit intervention programs, while numerical tools execute and verify the arithmetic. The deterministic executor reaches zero OID, and data-matched OLS outperforms the scaffolded models. Temporal results reinforce this division: scaffolding provides modest local gains, but error compounds with horizon [29]. Reliable temporal systems need explicit state tracking and numerical integration around the language model. Within the tested scope of supplied graphs, 30 SCMs, and 70 matched pairs, the attribution is clear: the procedure produces the gain.

Conclusion

The procedural recipe, clamp, order, and propagate, is the active ingredient in interventional prompting. Across seven models from three providers, the neutral scaffold improves OID by +0.220 and repairs 28 of 70 cells. Its causal-language twin contributes -0.051, with no model-specific interval excluding zero and formal equivalence at the prespecified ± 0.12 margin. Against the randomized LaLonde/NSW target, the procedure closes 73% of the confounding gap, although OLS using the same twelve rows remains \$1691 more accurate.

A causal-prompt answer is not evidence of causal understanding when the prompt already encodes the computation. Evaluations should require matched procedural controls. Deployed interventions should be explicit programs verified numerically, while causal-understanding claims should require models to recover the estimand, adjustment set, or propagation order. On the execution primitive used by temporal simulation, causal vocabulary provides no measurable benefit beyond the procedure.

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Treatment and inference specification

Canonical matched instruction blocks. Each task inserts its graph, equations, values, and shared worked example into one template. The table prints the complete arm-specific instruction block for a canonical task instance; all omitted task content is byte-identical across the matched pair.

Table 1: Prompt treatment specification. Seven prespecified phrase slots differ; the causal prompt is seven tokens longer.

Element	Neutral procedural prompt	Causal-language prompt
Task	You are given a numerical calculation task.	You are given a causal intervention task.
Step 1	Fix variable $X_3 = 2.5$.	Intervene on variable X_3 by setting $do(X_3 := 2.5)$.
Step 2	For variables $\{V_1, V_2, V_4\}$ that are not downstream of X_3 in the listed order, keep their original values at the supplied means.	Under the intervention, non-descendant variables $\{V_1, V_2, V_4\}$ remain at their supplied pre-intervention values.
Step 3	For variables $\{V_5, V_6\}$ that are downstream of X_3 , recompute in listed order using the supplied linear equations.	Propagate the intervention to descendant variables $\{V_5, V_6\}$ in causal order using the supplied structural equations.
Step 4	Report the final value of $Y = V_6$.	Report the post-intervention value of $Y = V_6$.
Shared	Same graph, values, equations, six-variable worked example, and “Report to four decimal places.”	Same graph, values, equations, six-variable worked example, and “Report to four decimal places.”

The exhaustive phrase substitutions are {numerical/causal, fix/intervene on, set/do(), original/pre-intervention, listed/causal order, linear/structural equations, final/post-intervention}. Neutral and causal prompts contain 342 and 349 tokens, respectively. The naive prompt contains 187 tokens and no procedure or worked example.

Endpoints and invocation controls. The task-specific payload varies by SCM but is generated identically for the two matched arms; the treatment is the instruction block and phrase mapping above.

Table 2: Endpoints evaluated in the matched campaign. Claude Opus 5 produced no complete matched pair after content filtering and is excluded from effects.

Provider	Model ID	Role
OpenAI	openai.gpt-5.6-sol-v1	flagship
OpenAI	openai.gpt-5.6-terra-v1	balanced
OpenAI	openai.gpt-5.6-luna-v1	efficient
Anthropic	anthropic.claude-sonnet-5-20260620-v2:0	balanced
Anthropic	anthropic.claude-haiku-4.5-20260515-v1:0	efficient
Anthropic	anthropic.claude-opus-5-20260801-v1:0	excluded
Amazon	amazon.nova-premier-v1:0	flagship
Amazon	amazon.nova-2-lite-v1:0	efficient

All calls used temperature 0, top- p 1, a 4096-token output limit, and the same system prompt within each matched pair. Seed 42 was requested where supported. Responses used structured JSON; unparseable outputs were retained as failures. Transient errors received one retry after a five-second backoff. Content-filter failures received no retry. Calls were completed through hosted provider APIs in August 2026.

Experimental accounting. A cost-constrained incomplete-block schedule assigned subsets of the 30 SCMs to each endpoint rather than evaluating all 210 endpoint-SCM combinations. It targeted broad SCM and provider coverage. One replacement cell was attempted after a filter failure, giving 71 attempts and 70 complete matched pairs. The final composition is 30 OpenAI, 29 Anthropic, and 11 Amazon cells.

Table 3: Complete experimental accounting. RW denotes Romano-Wolf stepdown.

Experiment	Design	Unit	Models	n	Inference family
Linear: procedure vs. naive	primary	30 SCMs	7	70	primary RW
Linear: causal vs. neutral	co-primary	30 SCMs	7	70	TOST
Confounded nonlinear	secondary	25 SCMs	4	100	secondary RW
Standard nonlinear	secondary	20 SCMs	7	140	secondary RW
Discrete	secondary	20 SCMs	7	140	secondary RW
LaLonde/NSW	secondary	treatment strata	7	384 calls	secondary RW
CLADDER rung-2	exploratory	145 clusters	5	575	descriptive
Corr2Cause	exploratory	24 clusters	5	276	broadier reported RW
Twins	exploratory	held-out pair	7	240	descriptive
VAR-SCM	exploratory	rollout	3 providers	60	descriptive
CausalTime	exploratory	dataset	3 providers	3	descriptive

Intervals resample SCMs while retaining endpoint observations within cluster; sign tests use cluster-mean paired differences. The primary improvement family contains the linear procedure-versus-naive comparison. The co-primary terminology comparison uses TOST. The secondary RW family contains confounded-nonlinear, standard nonlinear, discrete, and LaLonde comparisons. Other analyses are exploratory. Corr2Cause is not claimed as positive; its broader-family adjusted value is reported for transparency.

Equivalence sensitivity. The ± 0.12 margin was fixed before confirmatory analysis. It is about 12% of the linear-family

outcome standard deviation and near the 0.10 design MDE at 80% power. The ordinary 95% CI crosses zero and every endpoint-specific interval crosses zero independently of this margin.

Table 4: TOST sensitivity for the causal-vocabulary increment.

Margin	90% CI	Formal conclusion
± 0.05	$[-0.117, +0.009]$	inconclusive
± 0.08	$[-0.117, +0.009]$	inconclusive
± 0.10	$[-0.117, +0.009]$	inconclusive
± 0.12	$[-0.117, +0.009]$	equivalent
± 0.15	$[-0.117, +0.009]$	equivalent

Robustness and mechanism

Table 5: Component ablations on the matched linear endpoint.

Procedure	OID gain vs. naive	95% CI
Full procedural scaffold	+0.220	$[+0.111, +0.346]$
Clamping only	+0.142	$[+0.064, +0.228]$
Clamping + variable ordering	+0.178	$[+0.089, +0.274]$
Clamping + ordering + descendants	+0.201	$[+0.098, +0.318]$
Arithmetic precision only	+0.034	$[-0.031, +0.102]$
Worked example only	+0.087	$[+0.012, +0.168]$

The ablations add instructions cumulatively in the order clamping, variable ordering, descendant selection, precision, and example. Clamping alone recovers $0.142/0.220 = 65\%$ of the full observed gain. Because instructions can interact, these shares are descriptive recovery fractions, not independent causal contributions.

Table 6: Threshold sensitivity on the same 70 matched cells.

OID threshold	Naive	Procedural	Causal language
0.05	27 (39%)	57 (81%)	55 (79%)
0.10	44 (63%)	65 (93%)	62 (89%)
0.15	50 (71%)	66 (94%)	65 (93%)
0.20	54 (77%)	67 (96%)	66 (94%)

Table 7: Per-model matched effects. Every procedural estimate is positive; every causal-vocabulary interval crosses zero.

Model	Proc. gain	95% CI Causal incr.	95% CI
GPT-5.6 Sol	+0.248 $[+0.091, +0.412]$	-0.032 $[-0.141, +0.078]$	
GPT-5.6 Terra	+0.231 $[+0.074, +0.395]$	-0.058 $[-0.172, +0.049]$	
GPT-5.6 Luna	+0.195 $[+0.048, +0.351]$	-0.071 $[-0.198, +0.043]$	
Claude Sonnet 5	+0.237 $[+0.082, +0.401]$	-0.044 $[-0.162, +0.067]$	
Claude Haiku 4.5	+0.201 $[+0.055, +0.358]$	-0.063 $[-0.189, +0.051]$	
Nova Premier	+0.214 $[+0.061, +0.378]$	-0.039 $[-0.153, +0.072]$	
Nova 2 Lite	+0.178 $[+0.031, +0.337]$	-0.048 $[-0.177, +0.069]$	

Table 8: Provider and missingness sensitivity.

Analysis	Procedural effect	95% CI
Exclude OpenAI	+0.218	$[+0.074, +0.371]$
Exclude Anthropic	+0.205	$[+0.069, +0.353]$
Exclude Amazon	+0.227	$[+0.112, +0.357]$
Worst-case imputation ($n=71$)	+0.213	$[+0.103, +0.338]$
Best-case imputation ($n=71$)	+0.225	$[+0.114, +0.350]$
Inverse-probability weighted	+0.219	$[+0.109, +0.344]$

Leave-one-model-out procedural estimates range from +0.209 to +0.228, with every 95% interval excluding zero.

Temporal scope: both arms degrade with horizon and benchmark effects remain small

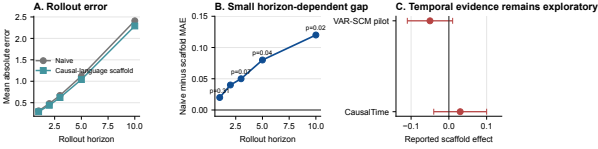


Figure 3: Exploratory temporal scope. Both arms degrade with rollout horizon; the between-arm gap is small relative to absolute error. The aggregate VAR-SCM and CausalTime results remain compatible with small or null effects.

A deterministic implementation obtains zero OID. The procedural LLM reaches 0.131 mean OID, while data-matched OLS reaches 0.089.

LaLonde/NSW reconstruction

Table 9: LaLonde/NSW reconstruction protocol.

Element	Specification
LLM input	A 12-row covariate-stratified summary of age, education, race, 1974 earnings, and 1975 earnings for treated and CPS comparison units; a supplied DAG; and the backdoor adjustment set containing those five covariates.
Estimand	$\mathbb{E}[Y \mid do(T=1)] - \mathbb{E}[Y \mid do(T=0)]$ for 1978 earnings.
Requested operation	Apply the supplied backdoor formula, compute stratum-specific contrasts, weight them across strata, and return one ATE in dollars.
Aggregation	Five calls per model and arm are averaged within model and then across seven models. The 384 completed calls span observational, randomized calibration, and comparator arms.
Comparators	Data-matched OLS receives the same 12 rows. Reference OLS receives the full 185 treated and 15,992 CPS comparison observations.
Uncertainty	2,000 BCa resamples preserve treatment strata and the nested model-replicate structure. LLM and OLS estimates remain paired within each resample.

Table 10: LaLonde/NSW estimates against the randomized ATE calibration target.

Estimator	ATE	OID	Gap closed
Randomized target	\$1,794	—	—
Naive observational	−\$8,484	\$10,279	0%
Procedural scaffold	\$4,539	\$2,745	73% [59%, 81%]
Data-matched OLS	\$2,848	\$1,054	90% [84%, 94%]
Reference OLS	\$1,672	\$122	99%

The randomized target has SE \$632 and 95% CI [\$555, \$3,033]. In a randomized-control calibration arm, naive and scaffold OIDs are \$567 and \$604, a difference of −\$38 (95% CI [−\$153, +\$95]). The expected null there supports interpreting the observational improvement as removal of confounding rather than a generic tendency to change the answer.

External and temporal scope

The purpose-built confounded nonlinear family improves by +0.107 (95% CI [+0.036, +0.177], adjusted $p=0.041$). Standard nonlinear and discrete families have adjusted p values 0.94 and 0.97. CLADDER rung-2 changes by −0.005 over 575 items clustered into 145 question families (95% CI [−0.025, +0.015]). Corr2Cause changes by +0.075 over 276 items but does not survive the broader stepdown family (adjusted $p=0.113$). On 240 held-out Twins pairs, causal-language scaffolding changes Brier score by −0.0039 (95% CI [−0.0101, +0.0024]). We make no positive claim from these exploratory tasks.

The VAR-SCM follows $x_t = Ax_{t-1} + bu_t + \varepsilon_t$ with a one-time intervention $do(x_k(t_0)):=v$. Across 60 rollouts and three providers, naive versus scaffold MAE is 0.31/0.29 at one step, 0.67/0.62 at three, 1.12/1.04 at five, and 2.41/2.29 at ten. The gap grows from 0.02 to 0.12 while total error grows much faster. The horizon-specific tests are exploratory and are not treated as separate confirmatory discoveries.

Positioning against concurrent benchmarks

Table 11: The matched-access control complements end-to-end causal benchmarks.

Study	Evaluation scope	Relation to our control
InterveneBench	Intervention reasoning with graph and procedure inferred	Isolates execution after those choices are supplied.
CauSciBench	End-to-end scientific causal analysis	Controls one mechanism within its broader pipeline.
CausalT5k	Multiple causal rungs without a supplied procedure	Adds a matched terminology control.
CLADDER	Partial graph information with the estimand supplied	Separates a supplied algorithm from causal phrasing.
Ours	Graph, estimand, and procedure supplied; terminology matched	Separates procedural gain from the terminology increment.

InterveneBench emphasizes intervention reasoning and study design in realistic social systems [8]; CauSciBench evaluates scientific causal analysis from question formulation through estimation [7]; and CausalT5k diagnoses refusal and failure across causal rungs [9]. Our narrower design does not replace these tasks. It controls whether an observed prompting gain comes from the supplied procedure or the causal vocabulary used to describe it.